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RESEARCH PAPER



Decoding the role of *OsPRX83* in enhancing osmotic stress tolerance in rice through ABA-dependent pathways and ROS scavenging

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ABSTRACT

Plant Class III peroxidases have diverse roles in controlling root hair growth, anther development, and abiotic and biotic stress responses. However, their abiotic stress response mechanism in rice remains elusive. Here, we identified a peroxidase precursor gene, *OsPRX83*, and investigated its role in enhancing osmotic stress tolerance in rice. We used *OsPRX83* overexpression and CRISPR-Cas9-generated mutant lines to elucidate *OsPRX83*'s function and expression patterns under stress conditions. The expression of *OsPRX83* was induced by H₂O₂, PEG, NaCl, and ABA treatments. Using qRT-PCR, RNA sequencing, and physiological assays, we demonstrated that overexpression of *OsPRX83* enhanced the osmotic and oxidative stress tolerance as compared to the wild-type and mutant seedlings, as evident from the higher survival rates, enhanced peroxidase (POD) and ascorbate peroxidase (APX) activities, and increased ABA sensitivity compared with mutants and wild-type plants. Transcriptome analysis further supported the involvement of *OsPRX83* in the ROS scavenging, by modulating the expression of *OsDREB1B*, *OsDREB1E*, *OsDREB1F*, *OsDREB1G* in response to osmotic treatment. In summary, our study suggests that *OsPRX83* plays a pivotal role in enhancing stress tolerance in rice through ABA-dependent pathways and ROS scavenging. Therefore, this study elucidates the function of a novel abiotic stress response gene in rice, thereby may contribute to a new genetic engineering resource for engineering drought-resistant rice varieties.

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Class III peroxidases; osmotic stress tolerance; oxidative stress tolerance; reactive oxygen species; rice

Introduction

Rice, a staple food for over half the global population¹, is highly vulnerable to drought stress due to its substantial water requirements, especially amid global climate change. Plants have developed various strategies, including morphological and physiological adaptations to counteract water deficiency and mitigate drought stress.^{2,3} Absciscic acid (ABA), a crucial plant hormone, is pivotal in responding to drought and salinity stresses.^{4,5} During drought stress, ABA-dependent and ABA-independent pathways activate diverse genes, including transcription factors. Many drought response genes are implicated in ABA-dependent pathways. Notably, the bZIP transcription factor OsABF1 directly regulated the expression of the protein phosphatase 2C (OsPP108 and OsPP48), forming a complex feedback loop through ABA signaling pathways.⁶ The PP2C and PYR1-like (PYL) proteins constituted ABA receptors in *Arabidopsis thaliana*.^{7,8} Overexpression of *OsPYL3* in *Arabidopsis* has been shown to enhance cold and drought stress tolerance.⁹ Multiple members of the cyclic nucleotide-gated channels (CNGC) family served as ABA-activated Ca²⁺ channels in *Arabidopsis* guard cells. Specifically, CNGC5, CNGC6, CNGC9, and CNGC12 have played crucial roles in the

ABA-specific patterns of cytosolic Ca²⁺ signaling pathways.¹⁰ To elucidate the mechanisms of ABA-induced regulation of CNGCs and Ca²⁺ signaling, researchers have discovered that OST1 physically interacted with CNGC5/6/9/12. Phosphorylation sites targeted by OST1 have been identified in CNGC5, CNGC6, CNGC9, and CNGC12. These phosphorylation sites are essential for ABA-induced activation of Ca²⁺ oscillations.¹¹

Plants respond to stress through intricate networks, one of which involves the participation of signal molecules such as reactive oxygen species (ROS) in the ABA signal pathways.¹² ROS exhibit a dual role in plant cells. They can be toxic byproducts of aerobic metabolism at high concentrations, but also toxic in a control manner during oxidative burst. At lower concentrations, they participate in signal transduction.¹³ To maintain cellular balance, plants have evolved diverse strategies for regulating ROS production, transport, and scavenging.¹⁴ Numerous proteins, notably superoxide dismutase (SOD), catalase (CAT), ascorbate peroxidase (APX), glutathione peroxidase (GPX), ascorbic acid, glutathione (GSH), and peroxidase (PRX), are integral to ROS detoxification.¹⁵ Among these, peroxidases are particularly crucial in oxidoreduction reactions, employing hydrogen peroxide (H₂O₂) as the electron acceptor in microorganisms, plants, and animals.¹⁶

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In the living organism, three peroxidase classes exist: Class I (ascorbate peroxidases), Class II (lignin peroxidases in certain fungi), and Class III (secretory peroxidases, Prxs).¹⁷ Among them, Class III peroxidases, specifically, contribute to stress tolerance,¹⁸ lignin biosynthesis,¹⁹ and root hair growth in *Arabidopsis*.²⁰ *Arabidopsis thaliana* and *Oryza sativa* possess 73 and 138 Prx genes, respectively,^{21,22} although the functions of most remain undetermined. In rice, MYBS1 regulated two Class III peroxidases by binding to the *bsr-d1* promoter with a single base change, enhancing resistance to *Magnaporthe oryzae*.²³ Additionally, *OsPrx30*, a Class III peroxidase in rice, localized in the endoplasmic reticulum, impacted rice resistance against *Xanthomonas oryzae* pv. *oryzae* (Xoo) when OsATH1 binded to its promoter, by scavenging ROS.²⁴ Furthermore, *OsPRX38*, another Class III peroxidase, improved arsenic tolerance by increasing antioxidant activities upon overexpression.²⁵

Despite Class III peroxidases having diverse roles in various biological processes, the molecular mechanisms of how Prxs respond to osmotic stress in rice remain unknown. Despite the size of this family, the functions of most members, particularly in response to abiotic stresses, remain largely unknown. Therefore, this study aimed to investigate the role of *OsPRX83*, a Class III peroxidase gene, in response to osmotic stress in rice. To achieve this objective, we used a combination of gene expression analysis and functional characterization through overexpression and knockout studies. Understanding the function of *OsPRX83* in osmotic stress response will provide new insights into the complex regulatory networks involved in abiotic stress tolerance in rice, which has important implications for the development of stress-tolerant rice varieties.

Materials and methods

Plasmid construction and rice transformation

The full-length coding region of *OsPRX83*, derived from *Oryza sativa japonica* cv. *Nipponbare* cDNA, was cloned into the pCXUN-FLAG vector²⁶ to generate *OsPRX83* overexpression (OE) constructs driven by the ubiquitin promoter. For the *OsPRX83* CRISPR/Cas9 vector, a sgRNA sequence was designed using the Rice Information GateWay (RIGW, http://rice.hzau.edu.cn/rice_rs3/).²⁷ This sequence was then cloned into the pU3-sgRNA vector and subsequently introduced into the pH-Ubi-cas9-7 vector via a gateway cloning LR reaction (ThermoFisher Scientific). For expression pattern analysis, the *OsPRX83* promoter region (2 kb fragment) was inserted into the pCXGUS-P vector, driving the β -glucuronidase (GUS) reporter gene expression. *Agrobacterium tumefaciens* EH105 received these vectors, which were then transferred into *Nipponbare* callus. Transgenic homozygous lines in the T3 generation were utilized for further study. Table S1 lists all primers used for plasmid construction.

Plant materials and stress treatments

This study utilized the well-established *Oryza sativa japonica* cv. *Nipponbare* as the study material.²⁸ To investigate

OsPRX83's expression profile under diverse abiotic stresses and phytohormone treatments, wild type (wt) four-leaf seedlings were subjected to diverse treatments, including cold (4°C), drought (20% PEG 4000, w/v), salt (150 mM NaCl), and oxidative (1% H₂O₂) stress.²⁹ Stressed leaf and shoot samples were collected at 0, 1, 3, 6, 12, and 24 h post-treatment, with additional samples for cold stress at 36 and 48 h. Phytohormone treatments involved soaking seedlings in 1/2 Murashige and Skoog (MS) medium containing 100 μ M of ABA, GA, MeJA, and IAA, respectively. Root samples were obtained at 0 h, 15 min, 1 h, 3 h, 6 h, 12 h, and 24 h.

To assess the osmotic and oxidative stress tolerance of rice seedlings, seeds of *OsPRX83*-OE, mutants and wt were initially grown on solid 1/2 MS medium for 3–4 days, then transferred to 1/2 MS medium (control), 120 mM and 150 mM mannitol-D, and 1 μ M and 3 μ M Methyl viologen (MV) for oxidative stress.³⁰ Following cultivation in a growth chamber for 10 days under a 16-h light/8-h dark cycle at 28°C, the shoot and root lengths were measured, and seedling phenotypes were documented.

For drought resistance analysis in *OsPRX83* transgenic plants, three-leaf seedlings of *OsPRX83* transgenic and wt plants (15 plants per line) were grown in large plastic boxes filled with well-mixed soil and cultivated in a growth chamber for two weeks. The water deprivation treatments ensued for 10–12 days until all seedling leaves were wilted, followed by a 20-day water recovery period to calculate survival rates. The ABA sensitivity assay was conducted³¹ by transferring germinated seeds of *OsPRX83* transgenic lines and wt to 1/2 MS medium containing 1 μ M ABA, 3 μ M ABA, or no ABA (control). After 10 days of cultivation, seedling phenotypes were recorded, and relative shoot height and root length were measured for all plants, with each test performed in triplicates.

Physiological measurements

The proline and chlorophyll contents were assessed pre- and post-treatment with 120 mM mannitol for three days. Leaf samples from *OsPRX83* transgenic lines and wt were obtained using the method previously described.³² For reactive oxygen detection, detached leaves from each genotype (minimum three plants per genotype) were immersed in a 0.5 mg/ml NBT staining solution. Following vacuum application for 30 min, the leaves underwent staining for 24 h. Subsequent decolorization involved 95% ethanol in an 80°C water bath until the leaves lost green coloration. Images were captured using a Leica M165FC stereoscope. Ascorbate peroxidase (APX) and peroxidase (POD) activities were quantified by Peroxidase (POD) Assay Kit (for plants) and Ascorbate Peroxidase (APX) Assay Kit according to the manufacturer's guidelines (Nanjing Jian Cheng Bioengineering Institute). The principle of the Plant Peroxidase (POD) Assay Kit is based on the catalytic reaction of peroxidase with hydrogen peroxide, the enzyme activity is determined by measuring the absorbance change at 420 nm. For the principle of the APX Assay Kit, APX catalyzes the reaction between ascorbic acid (ASA) and hydrogen peroxide (H₂O₂), converting ASA to monodehydroascorbic acid (MDASA). As AsA is oxidized, the absorbance value at 290 nm wavelength in the solution decreases. The ascorbate

peroxidase (APX) activity is determined by calculating the decrease of the absorbance change at 290 nm per unit time.

***β*-Glucuronidase (GUS) staining analysis**

Different tissues from pOsPRX83: GUS transgenic rice, including roots, stems, leaves, and spikelets were collected and processed using modified methods based on Jefferson.³³ The samples were initially soaked in 90% acetone for 30 minutes on ice, then washed three times in a buffer solution (2 mM potassium ferricyanide, 2 mM potassium ferrocyanide, 0.2% Triton-X-100 in 0.2 M PBS, pH 7.2) with gentle shaking for five minutes per wash. Following this, the samples underwent staining in a solution (buffer solution +20 mg/ml X-Gluc) for 48 h at 37°C. Finally, the stained samples were photographed using a stereoscope (Leica M165FC).

RNA extraction and quantitative real-time PCR

Total RNA was isolated from various tissues of OsPRX83 transgenic lines and wild-type rice using the RNA Extraction Kit (Promega). The Reverse Transcription System Kit (Promega) was then employed to synthesize first-strand cDNA. Quantitative RT-PCR (qRT-PCR) was performed on an ABI Step One Real-Time PCR system (Applied Biosystems) using the Hieff qPCR SYBR Green Master Mix (Yeast). The reactions were conducted under the following conditions: initial denaturation at 95°C for 5 minutes, followed by 40 cycles of denaturation at 95°C for 10 seconds and annealing/extension at 60°C for 30 seconds, culminating in the stage of melting curve analysis. Samples from the wild-type or untreated wild-type tissue were chosen as controls. The relative gene expression levels were calculated with the $2^{-\Delta\Delta C_t}$ method,³⁴ employing the rice *ubiquitin3* gene as an endogenous control.³⁵ All data analyses were conducted using GraphPad Prism 8.0.2. The primers used for gene expression analysis are listed in Table S1.

RNA sequencing analyses

At the four-leaf stage, OsPRX83 overexpression lines, mutants, and wt seedlings were treated with 120 mM mannitol for three days. Total RNA was extracted from the leaves before and after treatment, with three biological replicates per sample, using the previously described method. The purified RNA samples were sent to Gene Denovo Biotechnology Co., Ltd. (Guangzhou, China) for cDNA library preparation using the Illumina HiSeq 2500 platform. The clean reads were then aligned to the Rice Annotation Project Database (RAP-DB). Differentially expressed genes (DEGs) were performed by DESeq2³⁶ software and shown in the Table S2. They were identified with a threshold of log2 (fold change) ≥ 1 (False Discovery Rate – FDR < 0.05).³⁷ Gene Ontology (GO) enrichment and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analyses^{38,39} were performed based on the DEGs data using the website (<https://www.omicsmart.com/>). Certain DEGs that are upregulated in overexpression (OE) lines and downregulated in mutants are chosen and validated using qRT-PCR.

Statistical analysis

All data analyses were performed using GraphPad Prism 8.0.2. All datasets from the overexpression lines, mutants, and wild type (wt) were evaluated using IBM SPSS Statistics 19. Values at the $p < 0.05$ indicated statistical significance. Different Letters indicate significant differences at $p < 0.05$ by one-way analysis of Variance (ANOVA)⁴⁰ and Duncan's multiple range post hoc test.

Results

Expression patterns of OsPRX83

The expression levels of OsPRX83 in various tissues were examined using qRT-PCR to investigate its function. The results indicated high expression in roots, but low expression in leaves, stems, and spikelets (Figure 1(a)). This expression profile was corroborated by a GUS staining assay, which showed a strong GUS signal in roots (Figure 1a) and a relatively weak signal in stems, leaves, and spikelets (Figure 1b-e). To further explore the expression patterns under different abiotic stresses and phytohormone treatments, a time course assay was conducted using qRT-PCR. The expression level of OsPRX83 peaked at 24 h under cold treatment (Figure S1 a), but was down-regulated under heat treatment (Figure S1 b). In 1% H₂O₂ and ABA treatments, OsPRX83 expression significantly increased at 1 h, followed by a decrease (Figure S1 c, f). High salt treatment significantly induced OsPRX83 expression (Figure S1 d), while the expression pattern under PEG treatment mirrored that of oxidative treatment (Figure S1 e). GA treatment up-regulated OsPRX83 expression between 30 min and 3 h, followed by a decrease (Figure S1 g). Finally, under MeJA treatment, the expression level peaked at 1 h (Figure S1 h), remaining constant thereafter during the time course.

Molecular characterization of OsPRX83 transgenic plant

To examine OsPRX83's function under abiotic stress, we generated OsPRX83 mutants using the CRISPR/Cas9 genome editing system. During the T0 generation, DNA of shoot tissues of transgenic seedlings was extracted, then designed primers to sequence the CRISPR-Cas9 knockout sites by designing specific primers, compared the sequencing results with the wild type. Two homozygous mutant lines, *cas9-4B1* and *4C1*, each with 4 bp deletions, were selected (Figure 2(a)). Additionally, we generated OE transgenic lines using the ubiquitin promoter. qRT-PCR analysis revealed a significantly increased expression levels in OE-7 and OE-12 (32-fold and 218-fold, respectively) compared to the wt (Figure 2(b)). Subsequent investigations utilized T3 homozygous transgenic lines.

Overexpression of OsPRX83 improved osmotic stress tolerance in rice

To assess the potential role of OsPRX83 in osmotic stress response, seeds from the OsPRX83-OE transgenic line OE-12, *cas9-4C1* mutant, and wt were germinated on 1/2 MS solid medium supplemented with 0, 120 mM, and 150 mM mannitol for 10 days (Figure 2(c)). Subsequent measurements were taken of the

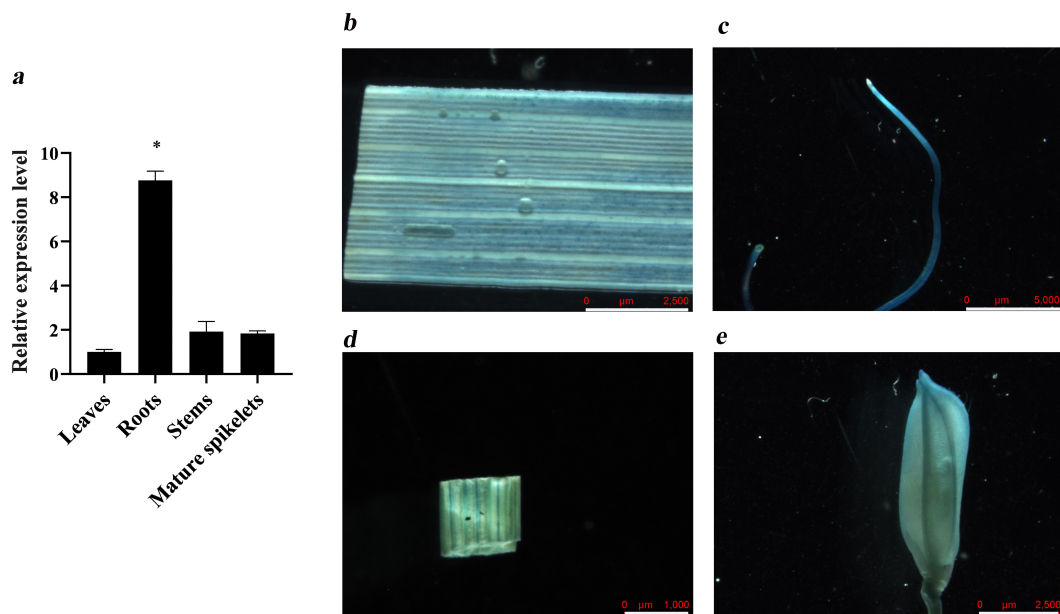


Figure 1. The tissue-specific expression profile of *OsPRX83*. (a) the relative expression levels of *OsPRX83* in various rice tissues (leaves, roots, stems, and mature spikelets). The expression level of the *OsPRX83* gene in leaves was normalized to 1, with *OsUPF3* representing the internal reference standard and the error bars ($\pm$ SE) representing three biological replicates. Asterisks indicate significant differences among leaf tissues of wild-type and other tissues of wild-type (Student's t-test, $p < 0.05$). (b-e) *OsPRX83* promoter-driven β -glucuronidase (GUS) expression pattern in (b) leaves, (c) roots, (d) stems and (e) flowers in mature spikelets.

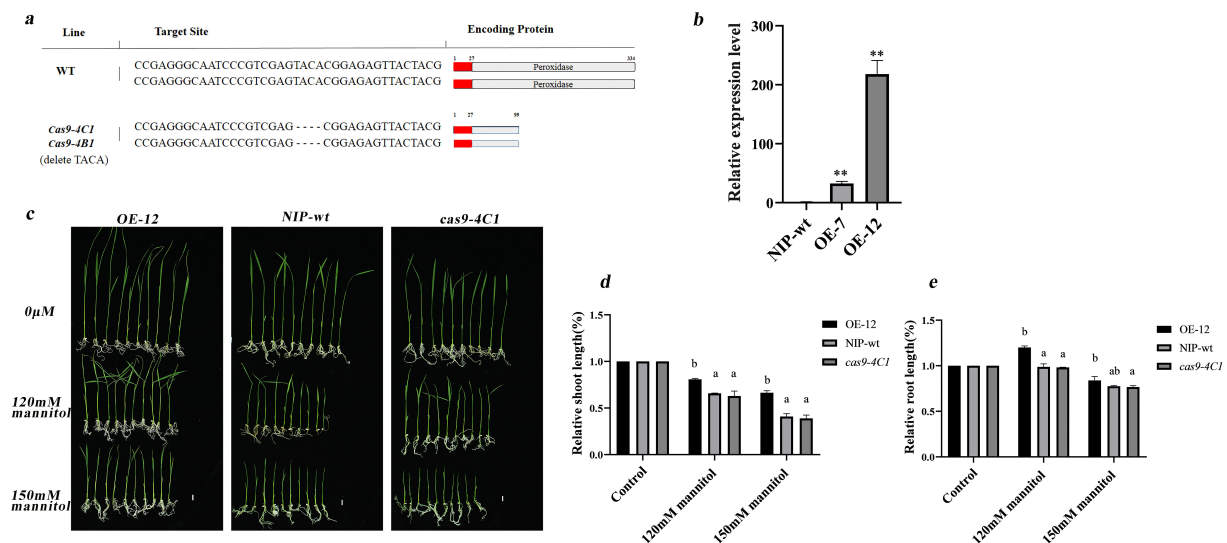


Figure 2. Phenotypic and molecular characterization of *OsPRX83* overexpression lines and mutants at the seedling stage. (a) analysis of mutation types and protein variation diagram in *Osgex3* mutant plants. Red boxes represent signal peptide (1-27 aa). (b) relative expression levels of *OsPRX83* in overexpression lines. Error bars represent standard error ($n = 3$). Asterisks indicate significant differences between wild-type and overexpression lines as per the Student's test (* $p < 0.05$, ** $p < 0.01$). (c) phenotypes of *OsPRX83* overexpressing lines OE-12, wild type, and mutants under osmotic stress (120 mM mannitol, 150 mM mannitol). Each genotype consists 20 seedlings for one test. Relative shoot length (d) and the relative root length (e) of *OsPRX83*-oe lines and wild type, and mutant. The data represent the mean $\pm$ SE ($n = 3$). Different letters represent the significant differences at $p < 0.05$.

seedlings' shoot height and root length. Figure 2(d) illustrates that, under osmotic stress, the OE-12 lines exhibited greater plant height than the wt and *cas9-4C1* mutant. Moreover, the root length of OE-12 was significantly greater than that of the wt and *cas9-4C1* mutant (Figure 2(e)). The experiment was replicated using the *OsPRX83*-OE transgenic line OE-7, mutant *cas9-4B1*, and wt, yielding similar results (Figure S2 a-c). Collectively, these findings suggest that *OsPRX83*-OE enhances rice seedling tolerance to osmotic stress during germination.

To validate *OsPRX83*'s role in osmotic stress response, we subjected 4-week-old seedlings of the OE line OE-12, wt, and mutant *cas9-4C1* to 120 mM mannitol for 3–4 days, followed by a 10-day recovery period (Figure 3(a)). Figure 3 b illustrated that OE-12 exhibited a superior survival rate (87%) compared to wt (60%) and mutant (30%). Physiological analyses demonstrated that the *OsPRX83*-OE lines accumulated elevated proline levels relative to the wt and mutant under stress conditions (Figure 3(c)). Additionally, post-treatment total chlorophyll

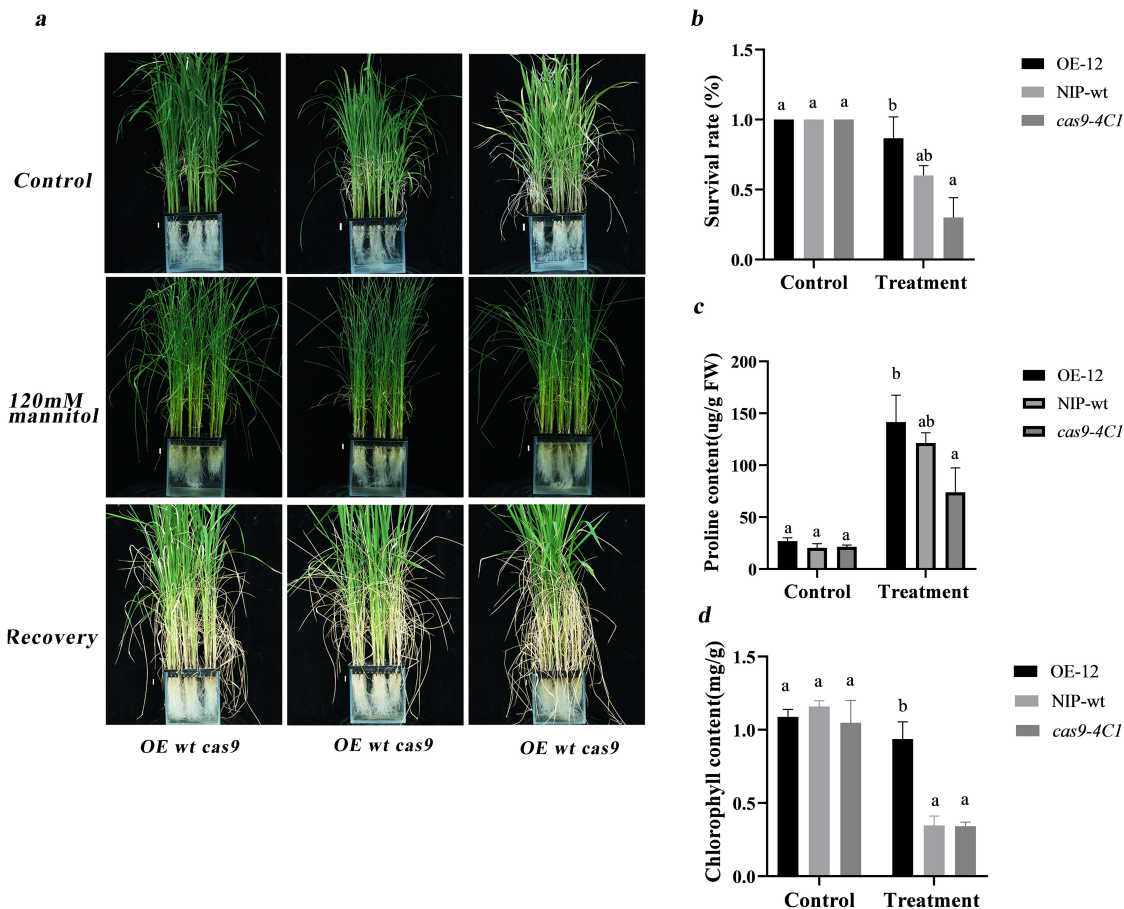


Figure 3. Phenotypes of *OsPRX83* overexpression transgenic line OE-12, wild type, and mutant *cas9-4C1* under osmotic stress. (a) *OsPRX83* transgenic plants (overexpression lines and mutants) and wt were subjected to a 120 mM mannitol solution for four days followed by a 10-day recovery period; scale bar = 1 cm. Each genotype consists 20 seedlings for one test. (b) survival rates of OE-12, wild type, and *cas9-4C1* seedlings were assessed before and after 120 mM mannitol treatment. (c) the proline content and total chlorophyll (d) were measured in the leaves of 4-week-old seedling before and after exposure to 120 mM mannitol. The data represent mean $\pm$ SE ($n = 3$). Different letters represent a significant difference at $p < 0.05$.

content significantly increased in OE lines versus wt and mutants (Figure 3(d)). Similar results emerged from stress treatment of OE line OE-7, wt, and mutant *cas9-4B1* (Figure S3). Thus, *OsPRX83*-OE augments osmotic stress tolerance in rice.

Overexpression of *OsPRX83* increased ABA sensitivity in rice

OsPRX83 expression significantly increased after a 15-min ABA treatment (Supplementary Figure S1 f). To probe *OsPRX83*'s role in ABA response, we conducted ABA sensitivity assays using *OsPRX83* transgenic lines, including OE lines (OE) and mutants. After a 10-day ABA treatment (0, 1, and 3 μ M), OE lines exhibited significantly suppressed seedling growth compared to wt and mutants (Figure 4(a), Figure S4 a), suggesting that *OsPRX83*-OE heightens ABA sensitivity during the seedling stage. Furthermore, *OsPRX83*-OE seedlings displayed significantly shorter relative shoot lengths than wt and mutants (Figure 4(b), Figure S4 b). These findings were corroborated by assessing relative root length (Figure 4(c) and S4 c).

To explore the potential link between *OsPRX83* expression and ABA response, we examined the expression levels of several ABA response genes via qRT-PCR. *OsPPI08* expression⁴¹

was significantly upregulated in *OsPRX83*-OE lines following ABA treatment for 30 min and 1 h, relative to wt and mutants (Figure S5 a). Additionally, *PYL3* expression⁹ peaked at 30 min in the OE lines before decreasing (Figure S5 b). Overall, these findings suggest that *OsPRX83* may participate in ABA signaling pathways by modulating *OsPPI08* and *PYL3* expression.

Overexpression of *OsPRX83* improved oxidative stress resistance in rice

OsPRX83 encodes a Class III peroxidase precursor, with expression induced by H_2O_2 within one hour (Fig. S1 c). To investigate *OsPRX83*'s role in oxidative stress response, we cultivated germinated seeds of *OsPRX83*-OE, wt, and mutant plants on 1/2 MS medium supplemented with 0, 1, or 3 μ M MV. After 10 days, MV treatment suppressed shoot growth in all seedlings, yet *OsPRX83*-OE plants displayed longer shoots than wt and mutants (Figure 5 (a) and S6 a). This observation was corroborated by relative shoot length measurements (Figure 5(b) and S6 b). Furthermore, the fresh weight per plant significantly declined post-treatment, but *OsPRX83*-OE plants showed less oxidative damage than wt and mutants (Figure 5(c) and S6 c). Thus, *OsPRX83* overexpression in rice conferred reduced growth inhibition and enhanced oxidative stress tolerance.

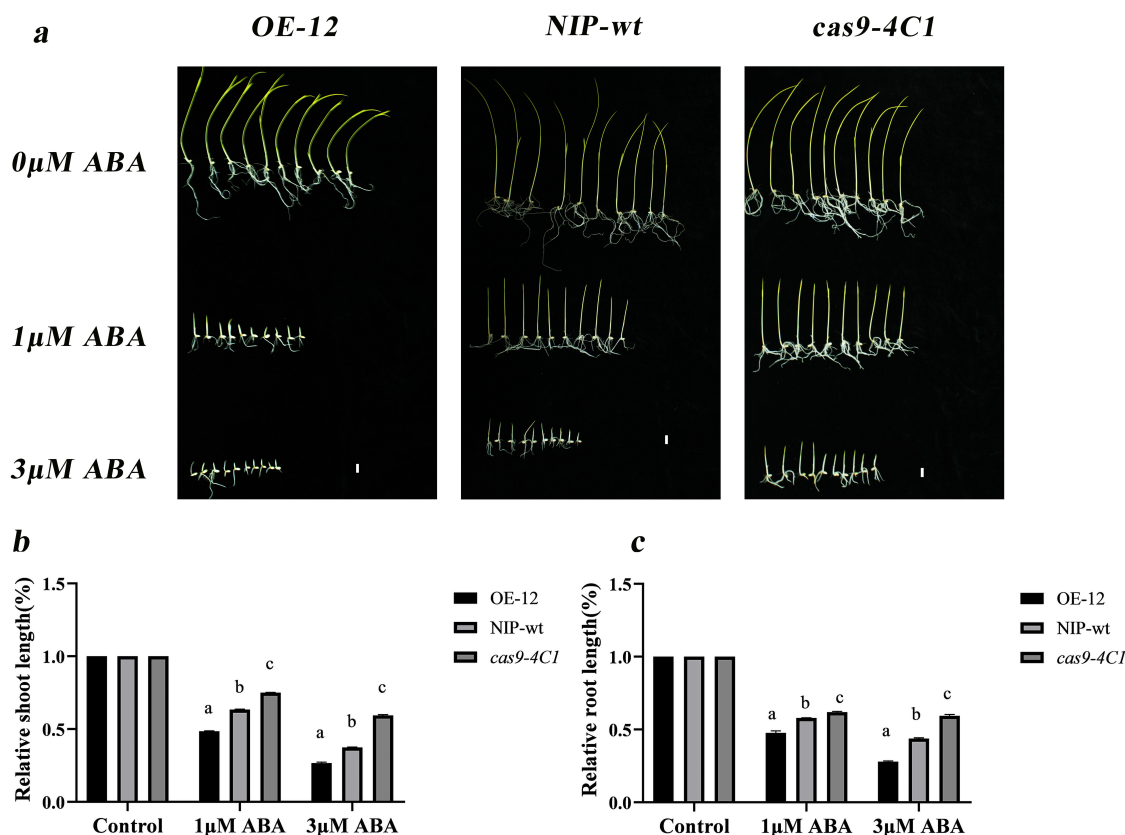


Figure 4. Phenotypes of *OsPRX83* overexpression lines OE-12, wild type, and mutant *cas9-4C1* under ABA treatment. (a) ABA sensitivity assay of *OsPRX83* overexpression line OE-12, wild type, and *cas9-4C1* mutant on 1/2 MS medium containing 0, 1, and 3 μM ABA (ABA treatments). Scale bar = 1 cm. (b) relative shoot length and (c) relative root length of OE-12 lines, *cas9-4C1* mutants, and wild type were measured 10 days after post-germination. Each genotype consists 20 seedlings for one test. Error bars represent the means $\pm$ SE ($n = 3$). Different letters indicate significant differences at $p < 0.05$.

Based on the aforementioned finding, it can be inferred that *OsPRX83* is associated with ROS metabolism. To validate this hypothesis, the qualitative nitroblue tetrazolium (NBT) staining assays were performed to examine the presence of superoxide anion before and after mannitol treatment. Upon mannitol treatment, *OsPRX83*-OE lines displayed fewer blue-stained leaf areas compared to wild type and mutants (Figure 6 a, b). Furthermore, peroxidase (POD) levels significantly increased in *OsPRX83*-OE lines after three days of 120 mM mannitol treatment (Figure 6(c)), as APX content (Figure 6 (d)). Collectively, these findings strongly support *OsPRX83*'s involvement in the ROS scavenging process.

Transcriptome profiling of *OsPRX83* transgenic lines

To further elucidate the molecular function of *OsPRX83*, a transcriptome analysis was performed on overexpression lines OE-12, wt, and *cas9-4C1* mutants.

Compared with the wild type, the overexpression line OE-12 displayed an upregulation of 604 genes and a down-regulation of 314 genes. Conversely, the *cas9-4C1* mutants revealed an upregulation of 923 genes and a down-regulation of 600 genes. Upon subjecting the OE-12 line to a three-day mannitol treatment, 5087 genes were up-regulated and 2469 genes were down-regulated, while the *cas9-4C1* mutants exhibited upregulation of 3349 genes and down-regulation of 1070 genes. Notably, a greater number of both up-regulated and

down-regulated genes were observed in OE-12 compared to *cas9-4C1* under mannitol treatment, as indicated by the volcano map (Figure 7(a)). Figure 7(b) illustrates the analysis of DEGs across the various lines. The comparison between the wt and the OE-12 line yielded 104 DEGs, while that between the wt and the *cas9-4C1* mutants resulted in 320 DEGs. Additionally, 564 DEGs were identified when comparing the wt and OE-12-T (under treatment), and 297 DEGs were found between the wt and *cas9-4C1*-T (under treatment). Among these DEGs, certain genes exhibited upregulation in OE-12 and down-regulation in *cas9-4C1*, with the wild type showing intermediate expression levels. A heat map (Figure 7(c)) was constructed to visualize the expression patterns of these selected DEGs. Additionally, the expression of four genes was confirmed using qRT-PCR (Figure 7(d)). Previous reports suggested that *OsDREB1B*, *OsDREB1E*, *OsDREB1F*, and *OsDREB1G*, which are implicated in abiotic stress responses,⁴² may have relevant biological functions. These transcriptome data suggest a potential role for *OsPRX83* in abiotic stress responses.

We conducted GO enrichment analysis and KEGG pathway analysis on DEGs using available data. Comparing the wt with the overexpression line OE-12, the top 20 molecular function GO terms emphasized oxidoreductase activity (GO: 0016491), dioxygenase activity (GO: 0051213), and carbon-oxygen lyase activity acting on phosphates (GO: 0016838). The predominant biological processes associated with these GO terms included

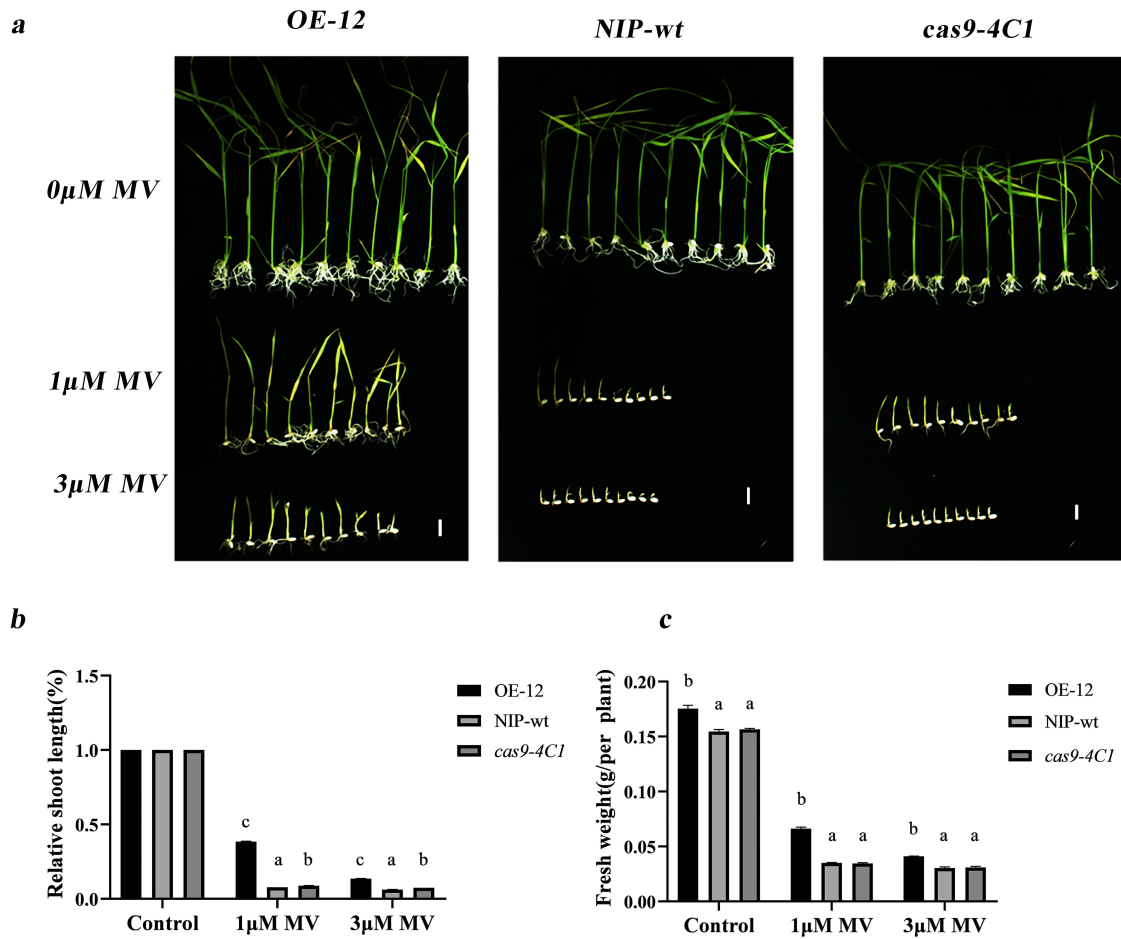


Figure 5. *OsPRX83* is involved in oxidative stress response. (a) enhanced tolerance of *OsPRX83* line OE-12 under oxidative stress using MV, as compared with wt and *cas9-4C1* mutants. Each genotype consists 20 seedlings for one test. Scale bar = 1 cm. (b) relative shoot length and (c) relative root length of *OsPRX83* transgenic lines and wild type. Data represent the means $\pm$ SE ($n = 3$). Different letters represent a significant difference at $p < 0.05$.

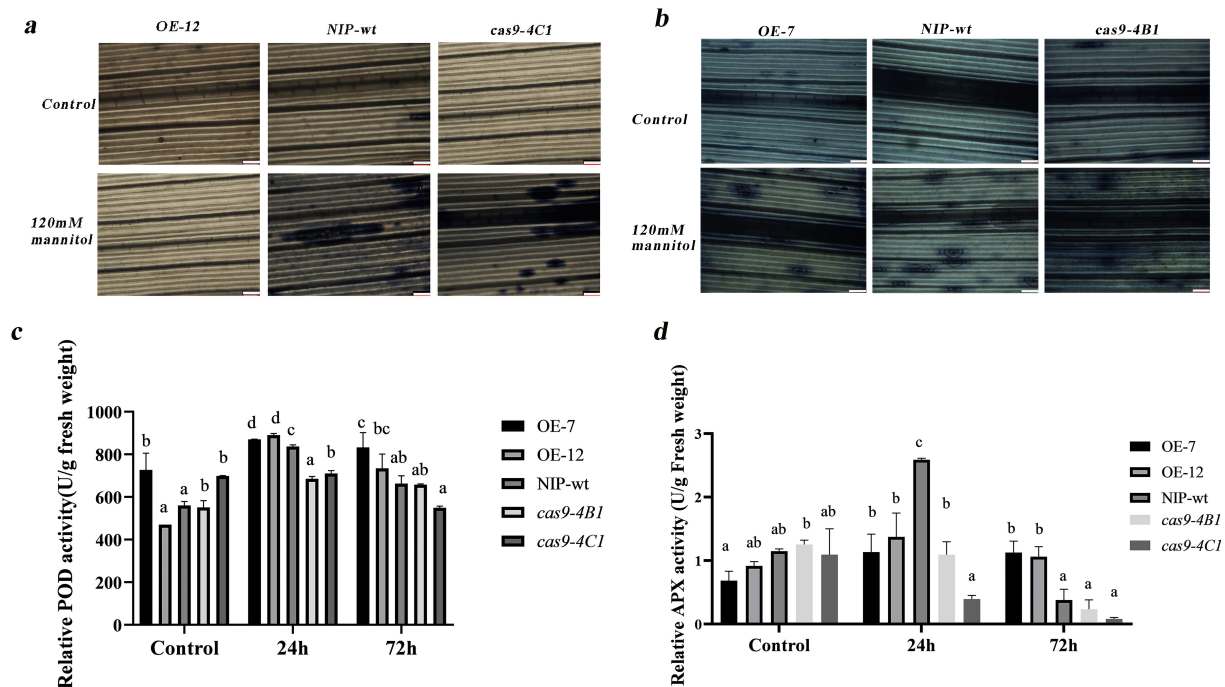


Figure 6. *OsPRX83* is involved in ROS scavenging. (a) nitroblue tetrazolium (NBT) staining assay using the leaves of *OsPRX83* overexpression lines OE-12, *cas9-4C1* mutants, and wt before and after 120 mM mannitol treatment, scale bar = 250 μ m. (b) NBT staining assay of *OsPRX83* overexpression lines OE-7, *cas9-4B1* mutants, and wild type before and after 120 mM mannitol treatment. Scale bar = 250 μ m. (c-d) changes in (c) peroxidase (POD) and (d) ascorbate peroxidase (APX) contents of the *OsPRX83* transgenic and wt leaves before (0 h) and after 120 mM mannitol treatment (24 h and 72 h). Different letters represent significant differences at $p < 0.05$.

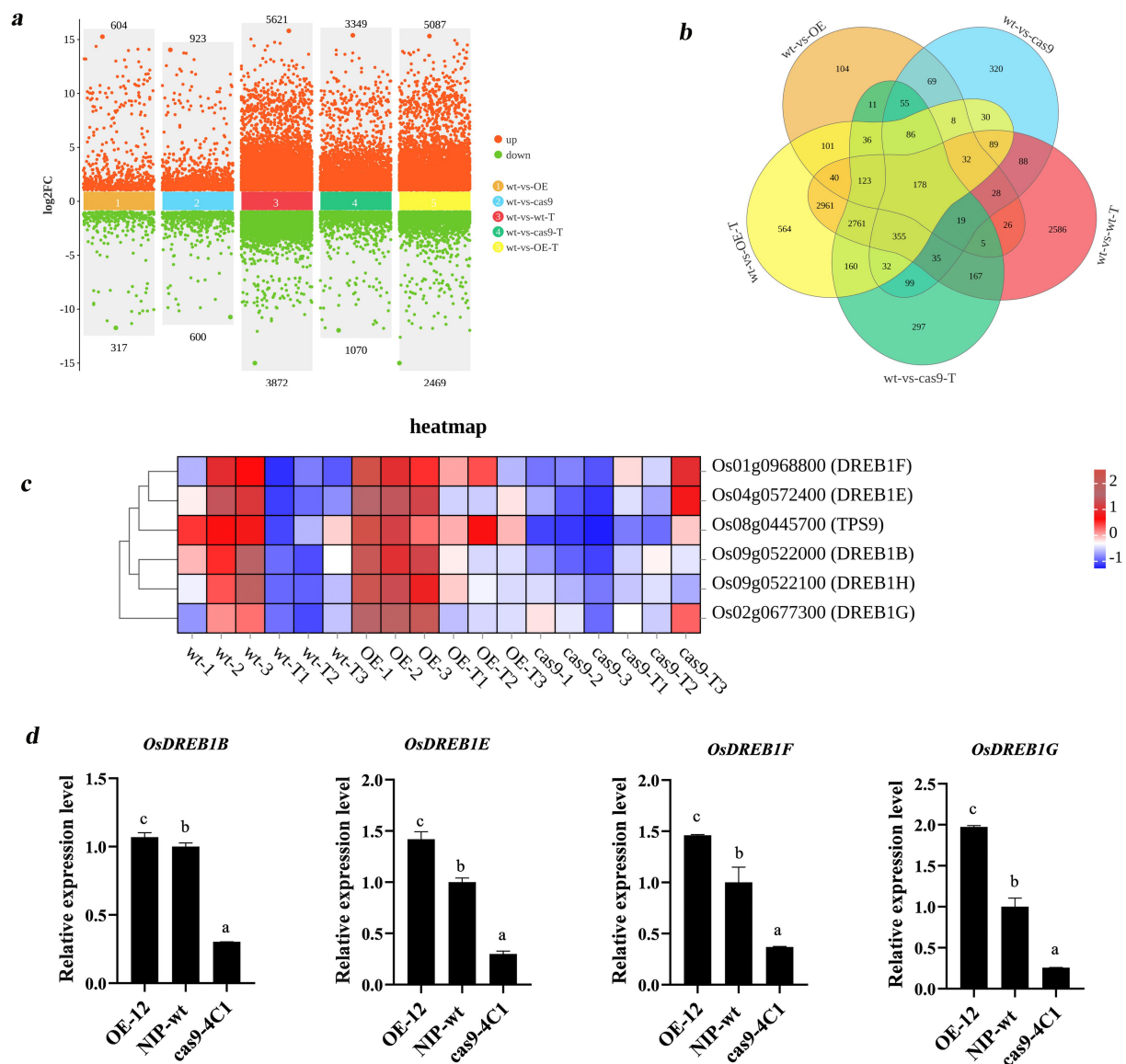


Figure 7. Transcriptome analysis of *OsPRX83* transgenic lines. (a) volcano plots depicting the number of up-regulated and down-regulated genes in the *OsPRX83*-oe line OE-12, *cas9-4C1* mutants, and wt. (b) venn diagrams illustrating the numbers of differentially expressed genes (DEGs) in OE vs wt, OE-T vs wt, cas9 vs wt, cas9-T vs wt, wt vs wt-T. (c) heat map shows the up-regulated gene expression in *OsPRX83*-oe transgenic lines compared with wt and mutants. (d) relative expression levels of DEGs in *OsPRX83* transgenic lines and wt plants as determined by qRT-PCR. The data represent the means $\pm$ SE ($n = 3$). Different numbers indicate significant differences at $p < 0.05$.

response to external stimulus, response to hormone, and reactive oxygen species metabolic process (GO: 0009605, 0009725, 0072593) (Figure S7 a). Upon treatment, the top 20 enriched biological process GO terms in the wt and OE-12-T comparison centered on plastid organization (GO: 0009657) and oxidoreduction coenzyme metabolic process (GO: 0006733) (Figure S7 b). These findings aligned with the GO terms related to cellular component, plastid part, plastid, plastid envelope, and plastid stroma (GO: 0044435, 0009536, 0009626, 0009532), which were highly enriched in this group. Conversely, the top 20 GO terms in both the wt vs *cas9-4C1* group and the wt vs *cas9-4C1*-T group (post-treatment) showed similarities, encompassing response to external stimulus, response to biotic stimulus, and response to stress (GO: 0050896, 0009607, 0006950) (Figure S7 c). Notably, the wt vs *cas9-4C1*-T group exhibited specific enrichment of terms related to response to abiotic stress,

response to oxidative stress, and response to reactive oxygen species (GO: 0009628, 0009679, 0000302) (Figure S7 d).

The MAPK signaling pathway in plants emerged consistently enriched across all groups in the KEGG pathway analysis. These groups include wt vs OE-12, wt vs OE-12-T, wt vs *cas9-4C1*, and wt vs *cas9-4C1*-T (Figure S8). This pathway, as previously reported, plays a pivotal role in plant responses to various abiotic stresses, with ROS serving as crucial signaling molecules.¹⁴ These findings, in conjunction with the results from the GO enrichment analysis, further substantiate the regulatory role of *OsPRX83* in ROS metabolism.

Discussion

Class III peroxidases (Prxs) are widely distributed plant-specific oxidoreductases with diverse roles in plant

development like control root hair growth²⁰ and promote root hair growth at low temperature⁴³ and abiotic stress responses.⁴⁴ However, the underlying molecular mechanism of their abiotic stress responses remains largely unknown. For example, overexpression of *AtPrx3* enhanced plant salt and drought tolerance.⁴⁵ Similarly, overexpression of two class III peroxidase genes (*CrPrx* and *CrPrx1*) in *Chrysanthemum* showed increased germination rates in *CrPrx1*-overexpressing lines as compared to wild type and *CrPrx* overexpressing lines under drought and salt stress conditions.⁴⁶ Overexpression of the *Arabidopsis* gene *AtPrx64* in tobacco also enhanced plant aluminum stress tolerance.⁴⁷ Overexpression of the *GsPRX9* in soybean was found to confer salt stress tolerance in transgenic seedlings.⁴⁸ Although the rice genome contains 138 Class III peroxidases, with two members, *OsPRX30* and *OsPRX38*, having reported functions.^{24,25} Phylogenetic analysis of the Prx family in rice is categorized into eight groups. *OsPRX30*, *OsPRX38*, *OsPRX83* belong to three subgroup of Prx family in rice, respectively.²⁴ Similarly, the roles of *OsPRX30* and *OsPRX38* are related with stresses responses, the function of *OsPRX83* may be associate with stress response as well.

In our study, the *OsPRX83* expression was induced by H₂O₂, NaCl, PEG, and ABA treatments, which led us to analyze its tolerance under these abiotic stress conditions using *OsPRX83* transgenic and wt plants. To further confirm our hypothesis, we found that *OsPRX83* overexpression enhanced tolerance to osmotic stress, via mannitol treatment, and oxidative stress, via MV treatment (Figures 3 and 5). Moreover, the *OsPRX83* overexpression augmented osmotic stress tolerance by boosting antioxidant reductase activities, specifically POD and APX (Figure 6). Therefore, we demonstrated *OsPRX83*'s positive regulatory role in enhancing osmotic stress resistance through enhanced ROS scavenging. Like *OsPRX30* and *OsPRX38*, *OsPRX83* is also considered as stress response gene through ROS metabolic pathways.

Moreover, the qRT-PCR analysis of RNA-seq data revealed the expression levels of *OsDREB1B*, *OsDREB1E*, *OsDREB1F*, and *OsDREB1G* (Figure 7d). These DREB transcription factors are known for their roles in abiotic stress response in rice. Specifically, *OsDREB1B* overexpression enhanced the cold and high salt stress tolerance. Moreover, *OsDREB1E* and *OsDREB1G* overexpression increased water deficit stress tolerance,⁴⁹ while *OsDREB1F* overexpression improved the tolerance against drought, salt, and cold stresses in rice and *Arabidopsis*.⁵⁰ Additionally, *OsATH1*, an AT-hook transcription factor, binded to the *OsPRX30* promoter, thereby regulating ROS accumulation.²⁴ Therefore, we hypothesize that *OsDREBs* may bind to the promoter of *OsPRX83*. To confirm this, *OsDREB1B*, *OsDREB1E*, *OsDREB1F*, and *OsDREB1G* are potential candidate genes for interaction with *OsPRX83*, requiring validation through yeast one-hybrid and Co-IP assays.

Also, *OsPRX83* (LOC_Os06g32990) also has a homologous gene (LOC_Os06g35490), which is also an unknown functional gene. Some genes have been found to interact with their homologous genes. The RNA binding proteins *OsLa1* and *OsLa2* are homologs in rice, and they interacted with together to regulate the pollen fertility and grain size.⁵¹ *OsPRX83* can also be interacted with its homologs in rice.

Compared with prior findings, our study elucidates the role of *OsPRX83* in modulating tolerance to osmotic stress through reactive oxygen species (ROS) scavenging pathways and abscisic acid (ABA) signaling pathways. The interacting proteins of *OsPRX83* implicated in this regulatory mechanism remains to be determined. Future investigations aim to unveil the intricate regulatory networks underlying the response to osmotic stress in rice, thereby enriching our understanding of the involvement of plant peroxidases in stress response mechanisms.

Conclusion

This study elucidates a novel Class III peroxidase gene in rice, *OsPRX83*. Utilizing overexpression and CRISPR-Cas9 knock-down mutants, we established *OsPRX83*'s role in responding to abiotic stresses. Our findings suggest that *OsPRX83* overexpression bolsters osmotic stress tolerance by augmenting ROS scavenging. Furthermore, *OsPRX83* positively regulates responses to oxidative stress and ABA sensitivity. This finding underscores the importance of ROS metabolism under abiotic stresses, thereby unveiling a new dimension in the functional analysis of plant peroxidases.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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Authors contribution statement

Yuchao Cui and Liang Chen supervised the project and designed the experiments. Han Bao designed the experiments. Han Bao performed most of the experiments, analyzed the data, and wrote the article. Xijun Zheng supported rice transgenic lines of *OsPRX83*. Chengke Luo assisted in the experiments. Yang Li provided practical assistance to Han Bao. All authors contributed to the article and approved the submitted version.

Abbreviations

APX	Ascorbate peroxidase
GUS	β-Glucuronidase
MDA	Malondialdehyde
MeJA	Jasmonic acid methyl ester
MV	Methylviologen
NBT	Nitroblue tetrazolium
<i>OsPRX83</i>	<i>Oryza sativa</i> peroxidase 83
PEG	Polyethylene glycol
PRX	Peroxidase
ROS	Reactive oxygen species
SOD	Superoxide dismutase

Data availability statement

All data supporting the findings of this study are available within the paper and within its supplementary materials published online. The datasets used in the study are available in online repositories (NCBI). The accession number can be found below: PRJNA1121593.

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